Jonathan Dray

Machine Learning Engineer · Deep Learning & RL · Marseille, France · open to US & Ireland

jonathandray83@gmail.com · linkedin.com/in/jonathan-d-049519221

PhD in machine learning (deep learning and reinforcement learning for real-time control), with a two-year ML engineering apprenticeship at Orange. Looking for a Machine Learning Engineer role.

Available from February 2027

Experience

PhD Researcher, ML for Adaptive Optics (CIFRE) — Since Dec. 2023

LAM (CNRS / AMU / CNES) & Bertin ALPAO — *Marseille, France*

Thesis: *Deep learning and reinforcement learning for adaptive optics in low Earth orbit optical downlinks*

- Developed a deep-learning wavefront reconstructor (CNN, Vision Transformer) robust to scintillation
- Showed in end-to-end simulation that reinforcement-learning control at **2 kHz** matches classical loops at **4 kHz**
- Deployed the RL controller in **C++ on GPU** inside the real-time control pipeline of an adaptive-optics testbed, running in closed loop at **2 kHz** (laboratory demonstration)
- Evaluated the PO4AO model-based RL controller under the non-stationary turbulence of LEO satellite tracking, and coupled it with the neural reconstructor
- Co-author of the first on-sky demonstration of PO4AO (PAPYRUS, Observatoire de Haute-Provence)

Machine Learning Engineer, apprenticeship — Sept. 2021 / Sept. 2023

Orange — *Marseille, France*

- Fine-tuned BERT-based models for enterprise text classification and improved their **GPU inference**
- Built supervised and unsupervised models for customer behaviour and business analytics
- Hyperparameter optimisation bringing **+15%** model performance
- Automated CI/CD pipelines (Docker, MLflow, cloud) for deployment and monitoring

Software Engineer, internship — 2020

NeuroMarseille / NeuroSchool — *Marseille, France*

- Full-stack development, statistical modelling and data analysis

Skills

GPU & ML	PyTorch, JAX, TensorFlow, GPU profiling, Vulkan, GLSL
Languages	Python, C++, C, SQL, MATLAB
Tools	Docker, MLflow, CI/CD, Linux, Git
Learning	CUDA, Triton, MLIR / LLVM, TVM
Spoken	French (native), English (fluent)

Education

PhD, Physics *Aix-Marseille Université* — 2023 / 2026

MSc, Mathematics & Data Science, with honours *Aix-Marseille Université* — 2021 / 2023

BSc, Mathematics *Aix-Marseille Université* — 2018 / 2021

Publications

2026 **J. Dray**, F. Quatresooz, M. Gray, B. Sinquin, J. Nousiainen, B. Neichel. Deep-learning-based reconstruction coupled with reinforcement learning for adaptive optics in LEO optical links. *Optical Engineering*, submitted (journal extension of SPIE 2026).

2026 **J. Dray**, K. Robert, C. Petit, B. Neichel. Numerical comparisons of VAR LQG-based predictive AO control and reinforcement-learning-based control methods for LEO-to-ground telecommunication links. *Proc. SPIE, Astronomical Telescopes + Instrumentation, Copenhagen.*

2026 **J. Dray**, M. Gray, B. Sinquin, et al. Deep-learning-based reconstruction coupled with reinforcement learning for adaptive optics in LEO optical links. *Proc. SPIE, Astronomical Telescopes + Instrumentation, Copenhagen.*

2026 J. Nousiainen, V. Chambouleyron, B. Neichel, . . . , **J. Dray**, et al. On-sky demonstration of reinforcement learning for adaptive optics control: PO4AO on PAPYRUS at OHP. *Accepted.* doi:10.48550/arXiv.2606.10771

2025 **J. Dray**, B. Sinquin, M. Gray, B. Neichel, et al. Towards AI-based predictive controllers for adaptive optics for LEO and GEO communications. *Proc. SPIE 13699, ICSO.*

2024 **J. Dray**, B. Sinquin, M. Gray, B. Neichel, et al. Comparison of predictive control laws in adaptive optics for free-space optical communications. *Proc. SPIE 13097, Adaptive Optics Systems IX.* doi:10.1117/12.3019728

2024 R. Fétick, T. Fusco, **J. Dray**, et al. PAPYRUS: from an AO system to an instrumental platform. *Proc. SPIE 13097.* doi:10.1117/12.3019758

2024 N. Levraud, N. Galland, H.-F. Raynaud, R. Fétick, **J. Dray**, et al. Simulation of LQG control with a pyramid wavefront sensor: preparation of PAPYRUS implementation. *Optica Imaging Congress, Toulouse.* doi:10.1364/AOPT.2024.OF1F.4

Talks & Posters

Talk *Towards AI-based predictive controllers for adaptive optics for LEO and GEO communications*
International Conference on Space Optics (ICSO), Antibes Juan-les-Pins, France — Oct. 2024

Talk *Deep-learning wavefront reconstruction with a Shack-Hartmann sensor under scintillation*
JRIOA, Adaptive Optics Research & Innovation Days (SFO), Orsay, France — May 2025

Poster *Comparison of predictive control laws in adaptive optics for free-space optical communications*
SPIE Astronomical Telescopes + Instrumentation, Yokohama, Japan — June 2024

Poster *Deep-learning-based reconstruction coupled with reinforcement learning for AO in LEO optical links*
SPIE Astronomical Telescopes + Instrumentation, Copenhagen, Denmark — July 2026